OBJECTIVE TYPE QUESTIONS

EXERCISE 1.6

1. If $n(A\times B)=6$ and $A=\{1,3\}$ then $n(B)$ is

(A) 1 (B) 2 (C) 3 (D) 6

Solution

$$n(A\times B)=n(A)\times n(B)$$

$$6=2\times n(B)\Rightarrow n(B)=3$$

[Option : (C)]

2. If $A=\{a,b,p\}, B=\{2,3\}, C=\{p,q,r,s\}$ then $n[(A\cup C)\times B]$ is

(A) 8 (B) 20 (C) 12 (D) 16

Solution

$$A\cup C=\{a,b,p,q,r,s\},\ n(A\cup C)=6,\ n(B)=2$$

$$n[(A\cup C)\times B]=n(A\cup C)\times n(B)=6\times 2=12$$

[Option : (C)]

3. If $A=\{1,2\}, B=\{1,2,3,4\}, C=\{5,6\}$ and $D=\{5,6,7,8\}$ then state which of the following statement is true?

(A) $(A\times C)\subset(B\times D)$ (B) $(B\times D)\subset(A\times C)$

(C) $(A\times B)\subset(A\times D)$ (D) $(D\times A)\subset(B\times A)$

Solution

$A\subset B$ and $C\subset D\Rightarrow A\times C\subset B\times D$

[Option : (A)]

4. If there are 1024 relations from a set $A=\{1,2,3,4,5\}$ to a set B, then the number of elements in B is

(A) 3 (B) 2 (C) 4 (D) 8

Solution

Total number of relations $=1024=2^{10}$

$$\Rightarrow\quad 2^{n(A)\times n(B)}=2^{10}$$

$$\Rightarrow\quad 2^{5\times n(B)}=2^{10}\Rightarrow 5\times n(B)=10\Rightarrow n(B)=2$$

[Option : (B)]

5. The range of the relation $R=\{(x,x^2)\mid x \text{ is a prime number less than } 13\}$ is

(A) $\{2,3,5,7\}$ (B) $\{2,3,5,7,11\}$ (C) $\{4,9,25,49,121\}$ (D) $\{1,4,9,25,49,121\}$

Solution

Domain of $R=\{2,3,5,7,11\}$,

$f(x)=x^2\Rightarrow f(2)=4, f(3)=9, f(5)=25, f(7)=49, f(11)=121$

Range of $R=\{4,9,25,49,121\}$

[Option : (C)]

6. If the ordered pairs $(a+2,4)$ and $(5,2a+b)$ are equal then (a,b) is

(A) $(2,-2)$ (B) $(5,1)$ (C) $(2,3)$ (D) $(3,-2)$

Solution

Given : $(a+2,4)=(5,2a+b)$

$$\Rightarrow a+2=5\Rightarrow a=3$$

[Option : (D)]

7. Let $n(A)=m$ and $n(B)=n$. Then the total number of non-empty relations that can be defined from A to B is

(A) m^n (B) n^m (C) $2^{mn}-1$ (D) 2^{mn}

Solution

Total number of relations from A to $B = 2^{mn}$

Total number of non-empty relations from A to $B = 2^{mn}-1$

[Option : (C)]

8. If $\{(a,8),(6,b)\}$ represents an identity function, then the values of a and b are respectively

(A) (8, 6) (B) (8, 8) (C) (6, 8) (D) (6, 6)

Solution

If f is an identity function, then $f(x)=x$.

$a=8,\ b=6$ **[Option : (A)]**

9. Let $A=\{1,2,3,4\}$ and $B=\{4,8,9,10\}$.

The function $f: A \to B$ given by $f=\{(1,4),(2,8),(3,9),(4,10)\}$ is

(A) a many-one function (B) an identity function

(C) an one-to-one function (D) an into function

Solution

Distinct elements of A have distinct images in B.

So, f is a one-to-one function. **[Option : (C)]**

10. If $f(x)=2x^2$ and $g(x)=\dfrac{1}{3x}$, then $f \circ g$ is

(A) $\dfrac{3}{2x^2}$ (B) $\dfrac{2}{3x^2}$ (C) $\dfrac{2}{9x^2}$ (D) $\dfrac{1}{6x^2}$

Solution

$$(f\circ g)(x)=f\left[g(x)\right]=f\left(\frac{1}{3x}\right)=2\left(\frac{1}{3x}\right)^2=\frac{2}{9x^2}$$ **[Option : (C)]**

11. If $f: A \to B$ is a bijective function and $n(B)=7$, then $n(A)$ is

(A) 7 (B) 49 (C) 1 (D) 14

Solution

If $f: A \to B$ is a bijective function, then $n(A)=n(B) \Rightarrow n(A)=7$ **[Option : (A)]**

12. Let f and g be two functions defined by $f=\{(0,1),(2,0),(3,-4),(4,2),(5,7)\}$, $g=\{(0,2),(1,0),(2,4),(-4,2),(7,0)\}$. Then the range of $f \circ g$ is

(A) $\{0,2,3,4,5\}$ (B) $\{-4,1,0,2,7\}$ (C) $\{1,2,3,4,5\}$ (D) $\{0,1,2\}$

Solution

$(f\circ g)(0)=f\left[g(0)\right]=f(2)=0$, $(f\circ g)(1)=f\left[g(1)\right]=f(0)=1$,

$(f\circ g)(2)=f\left[g(2)\right]=f(4)=2$, $(f\circ g)(-4)=f\left[g(-4)\right]=f(2)=0$

$(f\circ g)(7)=f\left[g(7)\right]=f(0)=1$.

Range of $f \circ g=\{0,1,2\}$ **[Option : (D)]**

13. If $f(x)=\sqrt{1+x^2}$ then

(A) $f(xy)=f(x)\cdot f(y)$ (B) $f(xy)\geq f(x)\cdot f(y)$

(C) $f(xy)\leq f(x)\cdot f(y)$ (D) None of these

Solution

$$f(xy)=\sqrt{1+(xy)^2}=\sqrt{1+x^2y^2} \quad \text{--- (1)}$$

$$f(x)\cdot f(y)=\sqrt{1+x^2}\sqrt{1+y^2}=\sqrt{1+x^2+y^2+x^2y^2} \quad \text{--- (2)}$$

(1) and (2) $\Rightarrow f(xy)\leq f(x)\cdot f(y)$

[Option : (C)]

14. If $g=\{(1,1),(2,3),(3,5),(4,7)\}$ is a function given by $g(x)=\alpha x+\beta$ then the values of α and β are

(A) $-1,2$ (B) $2,-1$ (C) $-1,-2$ (D) $1,2$

Solution

Given : $g(x)=\alpha x+\beta$,

$g(1)=1\Rightarrow \alpha+\beta=1$ --- (1)

$g(2)=3\Rightarrow 2\alpha+\beta=3$ --- (2)

Solving (1) and (2), we get $\alpha=2,\beta=-1$

[Option : (B)]

15. The function $f(x)=(x+1)^3-(x-1)^3$ is

(A) linear (B) cubic (C) reciprocal (D) quadratic

Solution

$$f(x)=(x+1)^3-(x-1)^3$$

$$=(x^3+3x^2+3x+1)-(x^3-3x^2+3x-1)=6x^2+2$$

$f(x)$ is a quadratic function .

[Option : (D)]

OBJECTIVE TYPE QUESTIONS

EXERCISE 2.10

1. For any two positive integers a and b, there exist unique integers q and r such that $a = bq + r$, where r must satisfy.

(A) $1 < r < b$ (B) $0 < r < b$ (C) $0 \le r < b$ (D) $0 < r \le b$

Solution $0 \le r < b$ **[Option : (C)]**

2. If the cube of any positive integer is divided by 9 then the possible remainders are

(A) 0,1,8 (B) 1,4,8 (C) 0,1,3 (D) 1,3,5

Solution

Let x be any positive integer. i.e., $x = 1,2,3,\cdots$ such that $x^3 \equiv r(\text{mod}9)$

When $x=1$, $1^3 = 1 \equiv 1(\text{mod}9)$; When $x=2$, $2^3 = 8 \equiv 8(\text{mod}9)$

When $x=3$, $3^3 = 27 \equiv 0(\text{mod}9)$; When $x=4$, $4^3 = 64 \equiv 1(\text{mod}9)$

When $x=5$, $5^3 = 125 \equiv 8(\text{mod}9)$; When $x=6$, $6^3 = 216 \equiv 0(\text{mod}9)\cdots$

[Option : (A)]

$\therefore$ The possible remainders are 0, 1, 8

3. If the H.C.F of 65 and 117 is $65m - 117$, then the value of m is

(A) 4 (B) 2 (C) 1 (D) 3

Solution

HCF of 65 and 117 is 13 **[Option : (B)]**

$65m - 117 = 13 \Rightarrow m = 2$

4. The sum of the exponents of the prime factors in the prime factorization of 1729 is

(A) 1 (B) 2 (C) 3 (D) 4

Solution

Prime factorization of 1729 $= 7^1 \times 13^1 \times 19^1$

Sum of the exponents of prime factors $= 1+1+1 = 3$ **[Option : (C)]**

5. The least number that is divisible by all the numbers from 1 to 10 is

(A) 2025 (B) 5220 (C) 5025 (D) 2520

Solution

Required number = LCM of 1,2,3,4,5,6,7,8,9,10

$= 2^3 \times 3^2 \times 5 \times 7$ (LCM by division method)

$= 2520$ **[Option : (D)]**

6. If k is a positive integer and $7^{4k} \equiv x(\text{mod}100)$ then x is

(A) 1 (B) 2 (C) 3 (D) 4

Solution

$7^{4k} = \left(7^4\right)^k = (2401)^k \equiv 1^k (\text{mod}100) = 1(\text{mod}100)$ **[Option : (A)]**

7. Given $F_1 = 1, F_2 = 3$ and $F_n = F_{n-1} + F_{n-2}$ then F_5 is

(A) 3 (B) 5 (C) 8 (D) 11

Solution

$F_3 = F_2 + F_1 = 3+1 = 4$; $F_4 = F_3 + F_2 = 4+3 = 7$; $F_5 = F_4 + F_3 = 7+4 = 11$ **[Option : (D)]**

8. The first term of an arithmetic progression is unity and the common difference is 4. Which of the following will be a term of this A.P.?

(A) 4551 (B) 10091 (C) 7881 (D) 13531

Solution

Since $a=1, d=4$, any term when divided by 4 gives 1

Here, 7881 is such a term **[Option : (C)]**

9. If 6 times of 6th term of an A.P is equal to 7 times the 7th term, then the 13th term of the A.P is

(A) 0 (B) 6 (C) 7 (D) 13

Solution

Given : $6t_6 = 7t_7 \Rightarrow 6(a+5d) = 7(a+6d) \Rightarrow a+12d = 0 \Rightarrow t_{13} = 0$ **[Option : (A)]**

10. An A.P consists of 31 terms. If its 16th term is m, then the sum of all the terms of this A.P. is

(A) $16m$ (B) $62m$ (C) $31m$ (D) $\frac{31}{2}m$

Solution

Given : $n=31$, $t_{16} = a+15d = m$

$S_{31} = \frac{31}{2}\left[2a+(31-1)d\right] \Rightarrow S_n = \frac{31}{2}\times 2(a+15d) \Rightarrow S_n = 31m$ **[Option : (C)]**

11. In an A.P., the first term is 1, the common difference is 4 and $S_n = 120$ then n is

(A) 6 (B) 7 (C) 8 (D) 9

Solution

$S_n = 120 \Rightarrow \frac{n}{2}\left[2+(n-1)4\right] = 120 \Rightarrow n\left[2n-1\right] = 120$

i.e., $2n^2 - n - 120 = 0$. Substitute the option values. $n=8$ satisfies. **[Option: (C)]**

12. If $A = 2^{65}$ and $B = 2^{64}+2^{63}+2^{62}+\cdots+2^0$ which of the following is true?

(A) B is 2^{64} more than A (B) A and B are equal

(C) B is larger than A by 1 (D) A is larger than B by 1

Solution

$B = 2^{64}+2^{63}+2^{62}+\cdots+2^0 = 2^0+2^1+\cdots+2^{64}$ is a G.P, where $a=1$, $r=2$, $n=65$

$B = S_{65} = \frac{1\left(2^{65}-1\right)}{2-1} = 2^{65}-1 = A-1 \Rightarrow A$ is larger than B by 1 **[Option : (D)]**

13. The next term of the sequence $\frac{3}{16}, \frac{1}{8}, \frac{1}{12}, \frac{1}{18}$ is

(A) $\frac{1}{24}$ (B) $\frac{1}{27}$ (C) $\frac{2}{3}$ (D) $\frac{1}{81}$

Solution

$\frac{t_2}{t_1} = \frac{1/8}{3/16} = \frac{2}{3}$, $\frac{t_3}{t_2} = \frac{1/12}{1/8} = \frac{2}{3} \Rightarrow$ The given sequence is a G.P with $r = \frac{2}{3}$.

$\therefore$ The next term $= \frac{1}{18}\times\frac{2}{3} = \frac{1}{27}$ **[Option : (B)]**

14. If the sequence $t_1, t_2, t_3, \cdots$ is an A.P. then the sequence $t_6, t_{12}, t_{18}, \cdots$ is

(A) a G.P. (B) an A.P.

(C) neither A.P. nor G.P. (D) a constant sequence

Solution

The suffice are in A.P. then the sequence is also in A.P. **[Option : (B)]**

15. The value of $(1^3+2^3+3^3+\cdots+15^3)-(1+2+3+\cdots+15)$ is

(A) 14400 (B) 14200 (C) 14280 (D) 14520

Solution

$$(1^3+2^3+3^3+\cdots+15^3)-(1+2+3+\cdots+15)=\left(\frac{15\times16}{2}\right)^2-\left(\frac{15\times16}{2}\right)=(120)^2-120=14280$$

[Option : (C)]

OBJECTIVE TYPE QUESTIONS

EXERCISE 3.20

1. A system of three linear equations in three variables is inconsistent if the corresponding planes

(A) intersect only at a point (B) intersect in a line

(C) coincide with each other (D) neither intersect in a line nor coincide

[Option : (D)]

Solution

2. The solution of the system $x+y-3z=-6, -7y+7z=7, 3z=9$ is

(A) $x=1,\ y=2,\ z=3$ (B) $x=-1,\ y=2,\ z=3$

(C) $x=-1,\ y=-2,\ z=3$ (D) $x=1,\ y=-2,\ z=3$

Solution

Substitute the option values in the equations and select the suitable option. **[Option : (A)]**

3. If $(x-6)$ is a factor of x^2-kx-6 then the value of k is

(A) 3 (B) 5 (C) 6 (D) 8

Solution **[Option : (B)]**

$\therefore P(6)=0 \Rightarrow 6^2-6k-6=0 \Rightarrow k=5$

4. $\dfrac{3y-3}{y} \div \dfrac{7y-7}{3y^2}$ is

(A) $\dfrac{9y}{7}$ (B) $\dfrac{9y^3}{(21y-21)}$ (C) $\dfrac{21y^2-42y+21}{3y^3}$ (D) $\dfrac{7(y^2-2y+1)}{y^2}$

Solution

$\dfrac{3y-3}{y} \div \dfrac{7y-7}{3y^2} = \dfrac{3y-3}{y} \times \dfrac{3y^2}{7y-7} = \dfrac{3(y-1)}{y} \times \dfrac{3y^2}{7(y-1)} = \dfrac{9y}{7}$ **[Option : (A)]**

5. $y^2+\dfrac{1}{y^2}$ is not equal to

(A) $\dfrac{y^4+1}{y^2}$ (B) $\left(y+\dfrac{1}{y}\right)^2$ (C) $\left(y-\dfrac{1}{y}\right)^2+2$ (D) $\left(y+\dfrac{1}{y}\right)^2-2$

Solution

$y^2+\dfrac{1}{y^2} \neq \left(y+\dfrac{1}{y}\right)^2$. **[Option : (B)]**

6. $\dfrac{x}{x^2-25} - \dfrac{8}{x^2+6x+5}$ gives

(A) $\dfrac{x^2-7x+40}{(x-5)(x+5)}$ (B) $\dfrac{x^2+7x+40}{(x-5)(x+5)(x+1)}$ (C) $\dfrac{x^2-7x+40}{(x^2-25)(x+1)}$ (D) $\dfrac{x^2+10}{(x^2-25)(x+1)}$

Solution

$$\frac{x}{x^2-25}-\frac{8}{x^2+6x+5}$$

$$=\frac{x}{(x+5)(x-5)}-\frac{8}{(x+1)(x+5)}=\frac{x(x+1)-8(x-5)}{(x+1)(x+5)(x-5)}=\frac{x^2-7x+40}{(x+1)(x^2-25)}$$

[Option : (C)]

7. The square root of $\frac{256x^8y^4z^{10}}{25x^6y^6z^6}$ is equal to

(A) $\left|\frac{16x^2z^4}{5y^2}\right|$ (B) $\left|\frac{16y^2}{x^2z^4}\right|$ (C) $\left|\frac{16y}{5xz^2}\right|$ (D) $\left|\frac{16xz^2}{5y}\right|$

Solution

$$\frac{256x^8y^4z^{10}}{25x^6y^6z^6}=\frac{256}{25}\frac{x^2z^4}{y^2}$$

$$\sqrt{\frac{256x^8y^4z^{10}}{25x^6y^6z^6}}=\frac{16}{5}\left|\frac{xz^2}{y}\right|$$

[Option : (D)]

8. Which of the following should be added to make x^4+64 a perfect square?

(A) $4x^2$ (B) $16x^2$ (C) $8x^2$ (D) $-8x^2$

Solution

$$x^4+64=\left(x^2+8\right)^2-16x^2$$

[Option : (B)]

9. The solution of $(2x-1)^2=9$ is equal to

(A) -1 only (B) 2 only (C) $-1, 2$ (D) None of these

Solution

$$(2x-1)^2=9\Rightarrow(2x-1)=\sqrt{9}\Rightarrow 2x-1=\pm 3\Rightarrow x=2, x=-1$$

or apply substitution method.

[Option : (C)]

10. If $4x^4-24x^3+76x^2+ax+b$ is a perfect square, the values of a and b are

(A) 100, 120 (B) 10, 12 (C) -120, 100 (D) 12, 10

Solution

$$\begin{array}{r|l} & 2x^2-6x+10 \\ \hline 2x^2 & 4x^4-24x^3+76x^2+ax+b \\ & 4x^4 \\ \hline 4x^2-6x & -24x^3+76x^2 \\ & -24x^3+36x^2 \\ \hline 4x^2-12x+10 & 40x^2+ax+b \\ & 40x^2-120x+100 \\ \hline & 0 \end{array}$$

Given polynomial is a perfect square if $ax+b=-120x+100$

$\therefore \quad a=-120, b=100$

[Option : (C)]

11. If the roots of the equation $q^2x^2+p^2x+r^2=0$ are the squares of the roots of the equation $qx^2+px+r=0$, then q, p, r are in

(A) A.P (B) G.P (C) Both A.P and G.P (D) none of these

Solution

Let α and β be the roots of $qx^2+px+r=0$. $\therefore \alpha+\beta=-\frac{p}{q}, \alpha\beta=\frac{r}{q}$

Now, α^2 and β^2 are the roots of $q^2x^2+p^2x+r^2=0$. $\therefore \alpha^2+\beta^2=-\frac{p^2}{q^2}, \alpha^2\beta^2=\frac{r^2}{q^2}$

$\alpha^2+\beta^2=-\frac{p^2}{q^2}<0$ which is not possible for any real numbers α, β, p, q.

We cannot proceed further.

12. Graph of a linear equation in two variables is a

(A) straight line (B) circle (C) parabola (D) hyperbola

[Option : (A)]

Solution Straight line

13. The graph $y=x^2+4x+4$ cuts the X-axis at

(A) no point (B) 1 point (C) 3 points (D) 2 points

Solution

Putting $y=0$, we get, $x^2+4x+4=0 \Rightarrow (x+2)(x+2)=0$

$\therefore x=-2,-2$ (twice). It cuts at one point $x=-2$. **[Option : (B)]**

14. For the given matrix $A=\begin{bmatrix}1 & 3 & 5 & 7\\ 2 & 4 & 6 & 8\\ 9 & 11 & 13 & 15\end{bmatrix}$ the order of the matrix A^T is

(A) 2×3 (B) 3×2 (C) 3×4 (D) 4×3

Solution

Order of $A=3\times4$. $\therefore$ Order of the matrix $A^T=4\times3$ **[Option : (D)]**

15. If A is a 2×3 matrix and B is a 3×4 matrix, the number of columns in AB is

(A) 3 (B) 4 (C) 2 (D) 5

Solution

Order of the matrix AB is 2×4. The matrix AB has 4 columns. **[Option : (B)]**

16. A matrix whose number of columns and rows are not equal, is

(A) a diagonal matrix (B) a rectangular matrix

(C) a square matrix (D) an identity matrix

Solution rectangular matrix **[Option : (B)]**

17. Transpose of a column matrix of order $1\times n$ is a

(A) unit matrix (B) diagonal matrix (C) column matrix (D) row matrix

Solution row matrix **[Option : (D)]**

18. Find the matrix X if $2X+\begin{bmatrix}1 & 3\\ 5 & 7\end{bmatrix}=\begin{bmatrix}5 & 7\\ 9 & 5\end{bmatrix}$.

(A) $\begin{bmatrix}-2 & -2\\ 2 & -1\end{bmatrix}$ (B) $\begin{bmatrix}2 & 2\\ 2 & -1\end{bmatrix}$ (C) $\begin{bmatrix}1 & 2\\ 2 & 2\end{bmatrix}$ (D) $\begin{bmatrix}2 & 1\\ 2 & 2\end{bmatrix}$

Solution

$$2X=\begin{bmatrix}5 & 7\\ 9 & 5\end{bmatrix}-\begin{bmatrix}1 & 3\\ 5 & 7\end{bmatrix}=\begin{bmatrix}4 & 4\\ 4 & -2\end{bmatrix}$$

$$\therefore X=\frac{1}{2}\begin{bmatrix}4 & 4\\ 4 & -2\end{bmatrix}=\begin{bmatrix}2 & 2\\ 2 & -1\end{bmatrix}$$

[Option : (B)]

19. If $A=\begin{bmatrix}1 & 2\\ 3 & 4\\ 5 & 6\end{bmatrix}$, $B=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$

which of the following, (i) A^2 (ii) B^2 (iii) AB (iv) BA can be calculated?

(A) (i) and (ii) only (B) (ii) and (iii) only (C) (ii) and (iv) only (D) all of these

Solution

Order of the matrix $A=3\times2$; Order of the matrix $B=3\times3$

(i) A^2 is not possible (ii) B^2 is possible (iii) AB is not possible (iv) BA is possible

[Option : (C)]

20. If $A=\begin{bmatrix}1 & 2 & 3\\ 3 & 2 & 1\end{bmatrix}$, $B=\begin{bmatrix}1 & 0\\ 2 & -1\\ 0 & 2\end{bmatrix}$ and $C=\begin{bmatrix}0 & 1\\ -2 & 5\end{bmatrix}$, which of the following statements are correct?

(i) $AB+C=\begin{bmatrix}5 & 5\\ 5 & 5\end{bmatrix}$

(ii) $BC=\begin{bmatrix}0 & 1\\ 2 & -3\\ -4 & 10\end{bmatrix}$

(iii) $BA+C=\begin{bmatrix}2 & 5\\ 3 & 0\end{bmatrix}$

(iv) $(AB)C=\begin{bmatrix}-8 & 20\\ -8 & 13\end{bmatrix}$

(A) (i) and (ii) only (B) (ii) and (iii) only (C) (iii) and (iv) only (D) all of these

Solution

(i) $AB+C=\begin{bmatrix}1 & 2 & 3\\ 3 & 2 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 2 & -1\\ 0 & 2\end{bmatrix}+\begin{bmatrix}0 & 1\\ -2 & 5\end{bmatrix}=\begin{bmatrix}5 & 4\\ 7 & 0\end{bmatrix}+\begin{bmatrix}0 & 1\\ -2 & 5\end{bmatrix}=\begin{bmatrix}5 & 5\\ 5 & 5\end{bmatrix}$ is correct.

(ii) $BC=\begin{bmatrix}1 & 0\\ 2 & -1\\ 0 & 2\end{bmatrix}\begin{bmatrix}0 & 1\\ -2 & 5\end{bmatrix}=\begin{bmatrix}0 & 1\\ 2 & -3\\ -4 & 10\end{bmatrix}$ is correct

(iii) Order of BA is 3×3. Order of C is 2×2. So, $BA+C$ is not possible.

From the above, one can conclude the option A

[Option : (A)]

OBJECTIVE TYPE QUESTIONS

EXERCISE 4.5

1. If the triangles ABC and EDF are similar, then

(A) $\angle B = \angle E$ (B) $\angle A = \angle D$ (C) $\angle B = \angle D$ (D) $\angle A = \angle F$

Solution

In ΔABC and ΔEDF, $\frac{AB}{DE} = \frac{BC}{FD}$.

By SAS criterion of similarity, $\Delta ABC \sim \Delta EDF$ if $\angle B = \angle D$ **[Option:(C)]**

2. In $\Delta LMN, \angle L = 60°, \angle M = 50°$. If $\Delta LMN \sim \Delta PQR$ then the value of $\angle R$ is

(A) $40°$ (B) $70°$ (C) $30°$ (D) $110°$

Solution

In ΔLMN, $\angle N = 180° - (\angle L + \angle M) = 70°$

$\angle R = \angle N = 70°$. **[Option:(B)]**

3. If ΔABC is an isosceles triangle with $\angle C = 90°$ and $AC = 5\,cm$, then AB is

(A) 2.5 cm (B) 5 cm (C) 10 cm (D) $5\sqrt{2}\,cm$

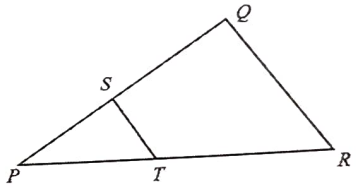
Solution

In ΔABC, $\angle C = 90°$ and $AC = 5\,cm$

ΔABC is isosceles $\Rightarrow BC = AC = 5\,cm$.

$AB^2 = BC^2 + CA^2 = 25 + 25 = 50$. $\therefore AB = \sqrt{50} = 5\sqrt{2}\,cm$ **[Option:(D)]**

4. In a given figure $ST \parallel QR, PS = 2\,cm$ and $SQ = 3\,cm$. Then the ratio of the area of ΔPQR to the area of ΔPST is



(A) 25 : 4 (B) 25 : 7

(C) 25 : 11 (D) 25 : 13

Solution

$PQ = PS + SQ = 2 + 3 = 5$

In ΔPQR and ΔPST, $ST \parallel QR \Rightarrow \Delta PQR \sim \Delta PST$

$\therefore \quad \frac{area\ of\ \Delta PQR}{area\ of\ \Delta PST} = \frac{PQ^2}{PS^2} = \frac{5^2}{2^2} = \frac{25}{4}$ **[Option:(A)]**

5. The perimeters of two similar triangles ΔABC and ΔPQR are 36 cm and 24 cm respectively. If $PQ = 10\,cm$, then the length of AB is

(A) $6\frac{2}{3}\,cm$ (B) $\frac{10\sqrt{6}}{3}\,cm$ (C) $66\frac{2}{3}\,cm$ (D) 15 cm

Solution Given : $\Delta ABC \sim \Delta PQR \Rightarrow \frac{AB}{PQ} = \frac{Perimeter\ of\ \Delta ABC}{Perimeter\ of\ \Delta PQR}$

$\frac{AB}{10} = \frac{36}{24} \Rightarrow AB = \frac{36}{24} \times 10 = 15\,cm$ **[Option:(D)]**

6. In the $\Delta ABC, DE \parallel BC, AB = 3.6\,cm$, $AC = 2.4\,cm$ and $AD = 2.1\,cm$. The length of AE is

(A) 1.4 cm (B) 1.8 cm (C) 1.2 cm (D) 1.05 cm

Solution

By BPT, we have $\frac{AB}{AD}=\frac{AC}{AE}\Rightarrow\frac{3.6}{2.1}=\frac{2.4}{AE}$

$AE=1.4\,cm$ **[Option:(A)]**

7. In a ΔABC, AD is the bisector of $\angle BAC$. If $AB=8\,cm$. $BD=6\,cm$ and $DC=3\,cm$, then the length of AC is

(A) 6 cm (B) 4 cm (C) 3 cm (D) 8 cm

Solution

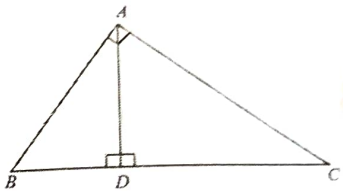
By angle bisector theorem, $\frac{AB}{AC}=\frac{BD}{DC}$

$\Rightarrow\frac{8}{AC}=\frac{6}{3}\Rightarrow AC=4\,cm$ **[Option:(B)]**

8. In the given figure $\angle BAC=90°$ and $AD\perp BC$ then

(A) $BD\cdot CD=BC^2$ (B) $AB\cdot AC=BC^2$

(C) $BD\cdot CD=AD^2$ (D) $AB\cdot AC=AD^2$



Solution

By a known result, $\Delta DBA\sim\Delta DAC$

$\therefore\frac{BD}{AD}=\frac{AD}{CD}\Rightarrow BD\cdot CD=AD^2$ **[Option:(C)]**

Remark : If AD is the perpendicular to the base BC from A then the triangles ABC, DBA and DAC are similar.

9. Two poles of heights $6m$ and $11m$ stand vertically on a ground. If the horizontal distance between them is $12m$, what is the distance between their tops?

(A) $13m$ (B) $14m$ (C) $15m$ (D) $12.8m$

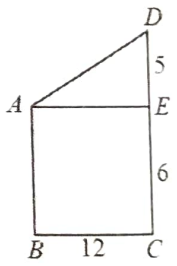
Solution

Let $AB=6\,m$, $CD=11\,m$

$\Rightarrow CE+ED=11\Rightarrow 6+ED=11\Rightarrow ED=5\,m$

In right angled triangle AED,

$AD^2=AE^2+DE^2=12^2+5^2=169\Rightarrow AD=13m$

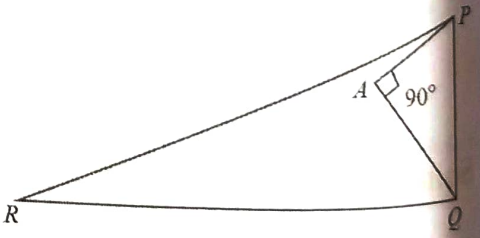


[Option:(A)]

10. In the given figure, $PR=26\,cm$, $QR=24\,cm$, $\angle PAQ=90°, PA=6\,cm$ and $QA=8\,cm$. Then $\angle PQR$ is

(A) 80° (B) 85°

(C) 75° (D) 90°



Solution

In the right angled ΔPAQ, $PQ=\sqrt{PA^2+QA^2}=\sqrt{6^2+8^2}=10$

In triangle ΔPQR, $PQ^2+QR^2=10^2+24^2=676=26^2=PR^2$.

$\Rightarrow\Delta PQR$ is a right angled triangle. $\therefore\angle PQR=90°$

[Option:(D)]

11. For a circle a tangent is perpendicular to the radius at the

(A) centre (B) point of contact (C) infinity (D) chord

Solution

A tangent is perpendicular to the radius at the point of contact. **[Option:(B)]**

12. The number of tangents can be drawn to the circle from an exterior point is

(A) one (B) two (C) infinite (D) zero

Solution

Twotangents can be drawn to the circle from an exterior point **[Option:(B)]**

13. The two tangents from an external points P to a circle with centre at O are PA and PB. If $\angle APB = 70°$ then the value of $\angle AOB$ is

(A) 100° (B) 110° (C) 120° (D) 130°

Solution

$OAPB$ is a quadrilateral.

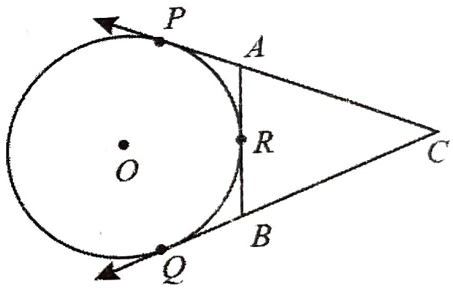
$\therefore \quad \angle AOB + \angle OBP + \angle BPA + \angle OAP = 360°$

$\Rightarrow \angle AOB + 90° + 70° + 90° = 360° \Rightarrow \angle AOB = 110°$ **[Option:(B)]**

14. In the given figure CP and CQ are tangents to a circle with centre at O. ARB is another tangent touching the circle at R. If $CP = 11\,cm$ and $BC = 7\,cm$, then the length of BR is

(A) 6 cm (B) 5 cm

(C) 8 cm (D) 4 cm



Solution

CP and CQ are tangents to a circle $\Rightarrow CP = CQ$

Now, $BC + BQ = CQ = 11 \Rightarrow BQ = 4\,cm$

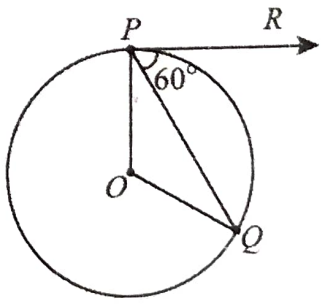
Given that ARB is another tangent touching the circle at R.

$\therefore BR = BQ = 4cm$ **[Option:(D)]**

15. In the given figure if PR is tangent to the circle at P and O is the centre of the circle, then $\angle POQ$ is

(A) 120° (B) 100°

(C) 110° (D) 90°



Solution

In ΔOPQ, $\angle OPQ = 90° - 60° = 30°$ and $OP = OQ$. Hence ΔOPQ is an isosceles triangle.

$\therefore \angle OPQ = \angle OQP = 30°$

Now, $\angle POQ = 180° - (\angle OPQ + \angle OQP) = 180° - 60° = 120°$ **[Option:(A)]**

OBJECTIVE TYPE QUESTIONS

EXERCISE 5.5

1. The area of triangle formed by the points $(-5,0),(0,-5)$ and $(5,0)$ is

(A) 0 sq.units (B) 25 sq.units (C) 5 sq.units (D) none of these

Solution

The area of the triangle **[Option : (B)]**

$= \frac{1}{2}\left|(25-0)+(0+25)+(0+0)\right| = \frac{1}{2}(50) = 25$

2. A man walks parallel to a wall, such that the distance between him and the wall is 10 units. Considering the wall as Y-axis, the path travelled by the man is

(A) $x=10$ (B) $y=10$ (C) $x=0$ (D) $y=0$

Solution

The path travelled by the man is parallel to Y - axis which is at a distance 10 units. **[Option : (A)]**

$\therefore$ The equation of the path is $x=10$.

3. The straight line $x=11$ is

(A) parallel to X-axis (B) parallel to Y-axis

(C) passing through the origin (D) passing through the point $(0,11)$

[Option : (B)]

Solution

$x=11$ is straight line parallel to Y - axis.

4. If $(5,7),(3,p)$ and $(6,6)$ are collinear, then the value of p is

(A) 3 (B) 6 (C) 9 (D) 12

Solution

Let the points be $A(5,7),B(3,p),C(6,6)$. Since A,B,C are collinear,

Slope of AB = Slope of $BC \Rightarrow \frac{p-7}{3-5} = \frac{6-p}{6-3}$ $\therefore p=9$ **[Option : (C)]**

5. The point of intersection of $3x-y=4$ and $x+y=8$ is

(A) $(5,3)$ (B) $(2,4)$ (C) $(3,5)$ (D) $(4,4)$

Solution

Solving the given equations, we get the point of intersection as $(3,5)$

(or) substitute and verify the points on the lines. **[Option : (C)]**

6. If the slope of the line joining $(12,3),(4,a)$ is $\frac{1}{8}$, the value of 'a' is

(A) 1 (B) 4 (C) -5 (D) 2

Solution

slope of the line joining $(12,3),(4,a) = \frac{1}{8}$

i.e., $\frac{a-3}{4-12} = \frac{1}{8} \Rightarrow \frac{a-3}{-8} = \frac{1}{8}$ $\therefore a=2$ **[Option : (D)]**

7. The slope of the line which is perpendicular to line joining the points $(0,0)$ and $(-8,8)$ is

(A) -1 (B) 1 (C) $\frac{1}{3}$ (D) -8

Solution

Slope of the line $m_1 = \frac{8-0}{-8-0} = -1$.

The slope of the line perpendicular to the line is $-\frac{1}{m_1} = 1$ **[Option : (B)]**

8. If the slope of a line is $\frac{1}{\sqrt{3}}$ then the slope of the perpendicular bisector of the line is

(A) $\sqrt{3}$ (B) $-\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) 0

Solution

slope of the line is $\frac{1}{\sqrt{3}}$.

The slope of the perpendicular bisector is $-\frac{1}{1/\sqrt{3}} = -\sqrt{3}$ **[Option : (B)]**

9. If A is a point on the y-axis whose coordinate is 8 and B is a point on the x-axis whose abscissa is 5 then the equation of the line AB is

(A) $8x+5y=40$ (B) $8x-5y=40$ (C) $x=8$ (D) $y=5$

Solution

x - intercept $a = 5$, y - intercept $b = 8$.

The equation of the line AB is $\frac{x}{a}+\frac{y}{b}=1 \Rightarrow \frac{x}{5}+\frac{y}{8}=1 \Rightarrow 8x+5y=40$ **[Option : (A)]**

10. The equation of a line passing through the origin and perpendicular to the line $7x-3y+4=0$ is

(A) $7x-3y+4=0$ (B) $3x-7y+4=0$ (C) $3x+7y=0$ (D) $7x-3y=0$

Solution

Since the required line passing through the origin $(0,0)$, check the points with the options. Either (C) or (D) is correct. But $3x+7y=0$ is perpendicular to the given line. **[Option : (C)]**

11. Consider four straight lines

(i) $l_1 : 3y = 4x+5$ (ii) $l_2 : 4y = 3x-1$ (iii) $l_3 : 4y+3x=7$ (iv) $l_4 : 4x+3y=2$

Which of the following statement is true?

(A) l_1 and l_2 are perpendicular (B) l_1 and l_4 are parallel

(C) l_2 and l_4 are perpendicular (D) l_2 and l_3 are parallel

Solution

Slope of $l_1 = \frac{4}{3}$, Slope of $l_2 = \frac{3}{4}$, Slope of $l_3 = -\frac{3}{4}$, Slope of $l_4 = -\frac{4}{3}$

$\Rightarrow l_2$ and l_4 are perpendicular. **[Option : (C)]**

12. For the straight line equation $8y = 4x + 21$, which of the following is true?

(A) The slope is 0.5 and the y-intercept is 2.6

(B) The slope is 5 and the y-intercepts is 1.6

(C) The slope is 0.5 and the y-intercept is 1.6

(D) The slope is 5 and y-intercept is 2.6

Solution

$$8y = 4x + 21 \Rightarrow y = \frac{4}{8}x + \frac{21}{8}$$

[Option : (A)]

Slope is 0.5, y-intercept $= 2.625 \simeq 2.6$

13. When proving that a quadrilateral is a trapezium, it is necessary to show

(A) Two sides are perpendicular

(B) One pair of opposite sides are parallel

(C) Opposite sides are parallel

(D) All sides are of equal length

[Option : (B)]

Solution It is enough to prove a pair of opposite sides are parallel

14. When proving that a quadrilateral is a parallelogram by using slopes you must find

(A) the slopes of two sides

(B) the slopes of two pairs of opposite sides

(C) the lengths of any two sides

(D) both the lengths and slopes of two sides

[Option : (B)]

Solution

15. $(2,1)$ is the point of intersection of two lines

(A) $x - y - 3 = 0 \; ; \; 3x - y - 7 = 0$

(B) $x + y = 3 \; ; \; 3x + y = 7$

(C) $3x + y = 3 \; ; \; x + y = 7$

(D) $x + 3y - 3 = 0 \; ; \; x - y - 7 = 0$

Solution

The point $(2,1)$ satisfies both the equations $x + y = 3$ and $3x + y = 7$. **[Option : (B)]**

OBJECTIVE TYPE QUESTIONS

EXERCISE 6.5

1. The value of $\sin^2\theta+\dfrac{1}{1+\tan^2\theta}$ is equal to

(A) $\tan^2\theta$ (B) 1 (C) $\cot^2\theta$ (D) 0

Solution

$$\sin^2\theta+\frac{1}{1+\tan^2\theta}=\sin^2\theta+\frac{1}{\sec^2\theta}=\sin^2\theta+\cos^2\theta=1$$

[Option : (B)]

2. $\tan\theta\,\text{cosec}^2\theta-\tan\theta$ is equal to

(A) $\sec\theta$ (B) $\cot^2\theta$ (C) $\sin\theta$ (D) $\cot\theta$

Solution

$$\tan\theta\,\text{cosec}^2\theta-\tan\theta=\tan\theta\left(\text{cosec}^2\theta-1\right)$$

$$=\tan\theta\times\cot^2\theta$$

$$=\left(\tan\theta\cot\theta\right)\cot\theta=\cot\theta$$

[Option : (D)]

3. If $\left(\sin\alpha+\text{cosec}\alpha\right)^2+\left(\cos\alpha+\sec\alpha\right)^2=k+\tan^2\alpha+\cot^2\alpha$, then the value of k is equal to

(A) 9 (B) 7 (C) 5 (D) 3

$$\left(\sin\alpha+\text{cosec}\alpha\right)^2+\left(\cos\alpha+\sec\alpha\right)^2$$

$$=\sin^2\alpha+\cos ec^2\alpha+2\sin\alpha\cos ec\alpha+\cos^2\alpha+\sec^2\alpha+2\cos\alpha\sec\alpha$$

$$=\left(\sin^2\alpha+\cos^2\alpha\right)+\cos ec^2\alpha+2+\sec^2\alpha+2$$

$$=1+\left(1+\cot^2\alpha\right)+2+\left(1+\tan^2\alpha\right)+2=7+\tan^2\alpha+\cot^2\alpha$$

$$\Rightarrow k=7$$

[Option : (B)]

4. If $\sin\theta+\cos\theta=a$ and $\sec\theta+\text{cosec}\theta=b$, then the value of $b(a^2-1)$ is equal to

(A) $2a$ (B) $3a$ (C) 0 (D) $2ab$

Solution

$$b\left(a^2-1\right)=\left(\sec\theta+\text{cosec}\theta\right)\left[\left(\sin\theta+\cos\theta\right)^2-1\right]$$

$$=\left(\sec\theta+\cos ec\theta\right)\left[\sin^2\theta+\cos^2\theta+2\sin\theta\cos\theta-1\right]$$

$$=\left(\frac{1}{\cos\theta}+\frac{1}{\sin\theta}\right)\left(2\sin\theta\cos\theta\right)$$

$$=2\left(\sin\theta+\cos\theta\right)=2a$$

[Option : (A)]

5. If $5x=\sec\theta$ and $\dfrac{5}{y}=\tan\theta$, then $x^2-\dfrac{1}{y^2}$ is equal to

(A) 25 (B) $\dfrac{1}{25}$ (C) 5 (D) 1

Solution

$$5x = \sec\theta \Rightarrow x = \frac{\sec\theta}{5}\ ; \qquad \frac{5}{y} = \tan\theta \Rightarrow \frac{1}{y} = \frac{\tan\theta}{5}$$

$$x^2 - \frac{1}{y^2} = \frac{\sec^2\theta}{25} - \frac{\tan^2\theta}{25} = \frac{\sec^2\theta - \tan^2\theta}{25} = \frac{1}{25}$$

[Option : (B)]

6. If $\sin\theta = \cos\theta$, then $2\tan^2\theta + \sin^2\theta - 1$ is equal to

(A) $\frac{-3}{2}$ (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $\frac{-2}{3}$

Solution

Given : $\sin\theta = \cos\theta \Rightarrow \theta = 45^\circ$

$$\therefore 2\tan^2\theta + \sin^2\theta - 1 = 2\tan^2 45^\circ + \sin^2 45^\circ - 1 = 2 + \frac{1}{2} - 1 = \frac{3}{2}$$

[Option : (B)]

7. If $x = a\tan\theta$ and $y = b\sec\theta$ then

(A) $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ (B) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (C) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (D) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$

Solution

$$x = a\tan\theta, y = b\sec\theta \Rightarrow \frac{x}{a} = \tan\theta, \frac{y}{b} = \sec\theta$$

$$sec^2\theta - \tan^2\theta = 1 \Rightarrow \frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$$

[Option : (A)]

8. $(1 + \tan\theta + \sec\theta)\ (1 + \cot\theta - \operatorname{cosec}\theta)$ is equal to

(A) 0 (B) 1 (C) 2 (D) -1

Solution

$$(1 + \tan\theta + \sec\theta)\ (1 + \cot\theta - \operatorname{cosec}\theta)$$

$$= \left[1 + \frac{\sin\theta}{\cos\theta} + \frac{1}{\cos\theta}\right]\left[1 + \frac{\cos\theta}{\sin\theta} - \frac{1}{\sin\theta}\right]$$

$$= \frac{\cos\theta + \sin\theta + 1}{\cos\theta} \times \frac{\sin\theta + \cos\theta - 1}{\sin\theta}$$

$$= \frac{(\sin\theta + \cos\theta)^2 - 1^2}{\cos\theta\sin\theta} = \frac{2\sin\theta\cos\theta}{\cos\theta\sin\theta} = 2$$

[Option : (C)]

9. If $a\cot\theta + b\operatorname{cosec}\theta = p$ and $b\cot\theta + a\operatorname{cosec}\theta = q$ then $p^2 - q^2$ is equal to

(A) $a^2 - b^2$ (B) $b^2 - a^2$ (C) $a^2 + b^2$ (D) $b - a$

Solution

$$p^2 - q^2 = (a\cot\theta + b\operatorname{cosec}\theta)^2 - (b\cot\theta + a\operatorname{cosec}\theta)^2$$

$$= a^2\cot^2\theta + b^2\operatorname{cosec}^2\theta + 2ab\cot\theta\operatorname{cosec}\theta - b^2\cot^2\theta - a^2\operatorname{cosec}^2\theta - 2ab\cot\theta\operatorname{cosec}\theta$$

$$= -a^2(\operatorname{cosec}^2\theta - \cot^2\theta) + b^2(\operatorname{cosec}^2\theta - \cot^2\theta) = b^2 - a^2$$

[Option : (B)]

10. If the ratio of the height of a tower and the length of its shadow is $\sqrt{3} : 1$, then the angle of elevation is

(A) 45° (B) 30° (C) 90° (D) 60°

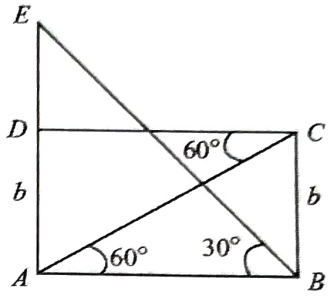
Solution

$$\frac{\text{height}}{\text{shadow}} = \sqrt{3} = \tan\theta \Rightarrow \theta = 60^\circ$$

[Option : (D)]

11. The electric pole subtends an angle of depression 30° at a point (B) on the same level as its foot. At another point 'b' metres above B, the depression of the foot of the tower is 60°. The height of the tower (in metres) is equal to

(A) $\sqrt{3}b$ (B) $\frac{b}{3}$ (C) $\frac{b}{2}$ (D) $\frac{b}{\sqrt{3}}$



$$\tan 60^\circ = \frac{b}{AB}$$

$$AB = \frac{b}{\sqrt{3}}$$

$$\tan 30^\circ = \frac{AE}{AB}$$

$$\frac{1}{\sqrt{3}} \times \frac{b}{\sqrt{3}} = AE \qquad \left(\because AB = \frac{b}{\sqrt{3}}\right)$$

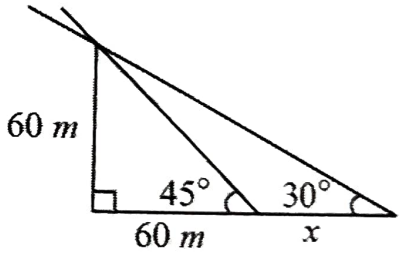
$$\therefore AE = \frac{b}{3}$$

height of the tower is $\frac{b}{3}$.

[Option : (B)]

12. A tower is 60 m heigh. Its shadow reduces by x metres,when the angle of elevation of the sun increases from 30° to 45°.Then x is equal to (assume that the edge of the shadow, top of the tower and the sun on the same line).

(A) 41.92 m (B) 43.92 m (43) 43 m (D) 45.6 m



$$\tan 30^\circ = \frac{60}{60+x}$$

$$\frac{1}{\sqrt{3}} = \frac{60}{60+x}$$

$$60+x = 60\sqrt{3}$$

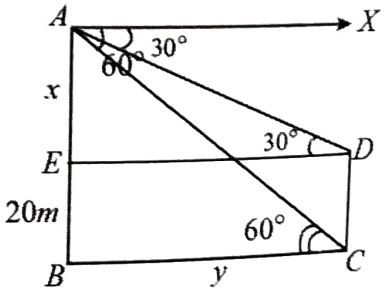
$$x = 60\left(\sqrt{3}-1\right) = 43.92 \ m$$

[Option : (B)]

13. The angles of depression of the top and bottom of 20 m tall building from the top of a multistoried building are 30° and 60° respectively. The height of the multistoried building and the distance between two buildings (in metres) are

(A) $20, 10\sqrt{3}$ (B) $30, 5\sqrt{3}$ (C) 20, 10 (D) $30, 10\sqrt{3}$

Solution



AB - multistoried building ; CD - tall building = 20m.

In ΔAED, $\tan 30^\circ = \frac{AE}{ED} = \frac{x}{y} \Rightarrow \frac{1}{\sqrt{3}} = \frac{x}{y}$ or $y = \sqrt{3}x$ ---(1)

In ΔABC, $\tan 60^\circ = \frac{AB}{BC} \Rightarrow \sqrt{3} = \frac{x+20}{y}$ or $y = \frac{x+20}{\sqrt{3}}$ --- (2)

From (1) and (2), we have $\sqrt{3}x = \frac{x+20}{\sqrt{3}} \Rightarrow x = 10m$.

When $x = 10m$, from (1), we get $y = 10\sqrt{3}m$

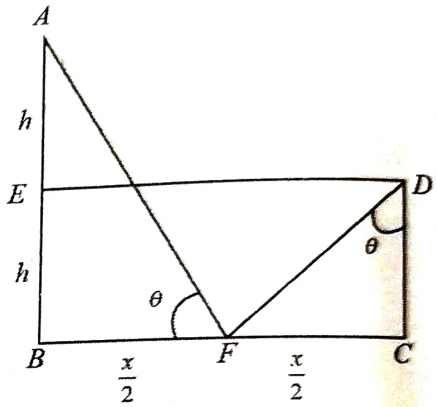
$\therefore$ Distance between two buildings $y = 10\sqrt{3}m$

Height of the multistoried building $x + 20 = 30m$ **[Option : (D)]**

14. Two persons are standing 'x' metres apart from each other and the height of the first person is double that of the other. If from the middle point of the line joining their feet an observer finds the angular elevations of their tops to be complementary, then the height of the shorter person (in metres) is

(A) $\sqrt{2}x$ (B) $\frac{x}{2\sqrt{2}}$ (C) $\frac{x}{\sqrt{2}}$ (D) $2x$

Solution



Let CD = Height of the second person in metres = h

AB = Height of the first person = $2h$. Let $BC = x$.

$\therefore BF = FC = \frac{x}{2}$

In ΔABF, $\tan\theta = \frac{AB}{BF} = \frac{2h}{\frac{x}{2}} = \frac{4h}{x}$ ---(1)

In right angled ΔDCF, $\tan\theta = \frac{FC}{DC} = \frac{\frac{x}{2}}{h} = \frac{x}{2h}$ ---(2)

From (1) and (2), we have $\frac{4h}{x} = \frac{x}{2h}$ or $8h^2 = x^2 \therefore h = \frac{x}{2\sqrt{2}}m$ **[Option : (B)]**

15. The angle of elevation of a cloud from a point $h$metres above a lake is β. The angle of depression of its reflection in the lake is $45°$. The height of location of the cloud from the lake is

(A) $\frac{h(1+\tan\beta)}{1-\tan\beta}$ (B) $\frac{h(1-\tan\beta)}{1+\tan\beta}$ (C) $h\tan(45°-\beta)$ (D) none of these

Solution

According to the problem 3 in the exercise 6.4, the height of of the location of the cloud from the lake is $\frac{h(\tan\theta_1 + \tan\theta_2)}{\tan\theta_2 - \tan\theta_1}$.

Here, $\theta_1 = \beta$. $\theta_2 = 45°$

Therefore the required height is $\frac{h(1+\tan\beta)}{1-\tan\beta}$.

[Option : (A)]

OBJECTIVE TYPE QUESTIONS

EXERCISE 7.5

1. The curved surface area of a right circular cone of height 15 *cm* and base diameter 16 *cm* is

(A) $60\pi cm^2$ (B) $68\pi cm^2$ (C) $120\pi cm^2$ (D) $136\pi cm^2$

Solution

Given : $h = 15cm,\ 2r = 16cm \Rightarrow r = 8cm.\ \ l = \sqrt{h^2 + r^2} = \sqrt{15^2 + 8^2} = 17cm.$

$\therefore$ CSA $= \pi rl = \pi \times 8 \times 17 = 136\pi cm^2$ **[Option : (D)]**

2. If two solid hemispheres of same base radius r units are joined together along their bases, then curved surface area of this new solid is

(A) $4\pi r^2$ sq.units (B) $6\pi r^2$ sq.units (C) $3\pi r^2$ sq.units (D) $8\pi r^2$ sq.units

Solution

If we join two solid hemispheres of same base radius r along their bases, we get a solid sphere of radius r units. Hence the CSA is $4\pi r^2$ Sq.units. **[Option : (A)]**

3. The height of a right circular cone whose radius is 5 *cm* and slant height is 13 *cm* is

(A) 12 *cm* (B) 10 *cm* (C) 13 *cm* (D) 5 *cm*

Solution

$h = \sqrt{l^2 - r^2} = \sqrt{13^2 - 5^2} = \sqrt{144} = 12cm$ **[Option : (A)]**

4. If the radius of the base of a right circular cylinder is halved keeping the same height, then the ratio of the volume of the two cylinders is

(A) 1:2 (B) 1:4 (C) 1:6 (D) 1:8

Solution

Radius of new cylinder $R = \frac{r}{2}$.

$$\frac{\text{Volume of new cylinder}}{\text{Volume of original cylinder}} = \frac{\pi R^2 h}{\pi r^2 h} = \frac{r^2/4}{r^2} = \frac{1}{4}$$ **[Option : (B)]**

5. The total surface area of a cylinder whose radius is $\frac{1}{3}$ of its height h is

(A) $\frac{9\pi h^2}{8}$ sq.units (B) $24\pi h^2$ sq.units

(C) $\frac{8\pi h^2}{9}$ sq.units (D) $\frac{56\pi h^2}{9}$ sq.units

Solution

Given : $r = \frac{h}{3}$. TSA $= 2\pi r(h + r) = 2\pi\left(\frac{h}{3}\right)\left(h + \frac{h}{3}\right) = \frac{8\pi h^2}{9}$ **[Option : (C)]**

6. In a hollow cylinder, the sum of the external and internal radii is 14 *cm* and the width is 4 *cm*. If its height is 20 *cm*, the volume of the material is

(A) $5600\pi\ cm^3$ (B) $1120\pi\ cm^3$ (C) $56\pi\ cm^3$ (D) $3600\pi\ cm^3$

Solution Given : $R + r = 14,\ w = R - r = 4,\ h = 20$

Volume of hollow cylinder $= \pi h(R^2 - r^2) = \pi h(R + r)(R - r)$

$= \pi \times 20 \times 14 \times 4 = 1120\pi cm^3$

[Option : (B)]

7. The radius of the base of a cone is tripled and the height is doubled then the volume increases to

(A) 6 times (B) 18 times (C) 12 times (D) unchanged

Solution

Let R and H be the radius and height of the new cone.

$R = 3r$ and $H = 2h$.

Volume of the new cone $= \frac{1}{3}\pi R^2 H = \frac{1}{3}\pi \cdot 9r^2 \cdot 2h = 18\left(\frac{1}{3}\pi r^2 h\right) = 18$ times **[Option : (B)]**

8. The total surface area of a hemi-sphere is how much times the square of its radius?

(A) π (B) 4π (C) 3π (D) 2π

Solution

TSA $= 3\pi r^2 \Rightarrow$ TSA is 3π times the square of its radius. **[Option : (C)]**

9. A solid sphere of radius x *cm* is melted and cast into a shape of a solid cone of same radius. The height of the cone is

(A) $3x\,cm$ (B) $x\,cm$ (C) $4x\,cm$ (D) $2x\,cm$

Solution

Here $r = x$, height $= h$.

Volume of Cone = Volume of sphere

$\Rightarrow \frac{1}{3}\pi x^2 h = \frac{4}{3}\pi x^3 \Rightarrow h = 4x\,cm$ **[Option : (C)]**

10. A frustum of a right circular cone is of height 16 *cm* with radii of its ends as 8 *cm* and 20 *cm*. Then, the volume of the frustum is

(A) $3328\pi\ cm^3$ (B) $3228\pi\ cm^3$ (C) $3240\pi\ cm^3$ (D) $3340\pi\ cm^3$

Solution

Given : $h = 16cm,\ R = 20cm,\ r = 8cm$. Volume of frustum $= \frac{1}{3}\pi h\left(R^2 + r^2 + Rr\right)$

$= \frac{1}{3}\pi \times 16 \times \left[20^2 + 8^2 + (20)(8)\right] = 3328\pi cm^3$ **[Option : (A)]**

11. A shuttle cock used for playing badminton has the shape of the combination of

(A) asylinder and a sphere (B) a hemisphere and a cone

(C) a sphere and a cone (D) frustum of a cone and a hemisphere

Solution

[Option : (D)]

12. A spherical ball of radius r_1 units is melted to make 8 new identical balls each of radius r_2 units. Then $r_1 : r_2$ is

(A) 2:1 (B) 1:2 (C) 4:1 (D) 1:4

Solution

$$\left.\begin{matrix}\text{Volume of a spherical ball}\\ \text{of radius } r_1\end{matrix}\right\} = 8\left\{\begin{matrix}\text{Volume of a ball}\\ \text{of radius } r_2\end{matrix}\right.$$

$\Rightarrow \frac{4}{3}\pi r_1^3 = 8 \times \frac{4}{3}\pi \times r_2^3 \Rightarrow \frac{r_1^3}{r_2^3} = 8$ or $\left(\frac{r_1}{r_2}\right)^3 = \left(\frac{2}{1}\right)^3 \therefore \frac{r_1}{r_2} = \frac{2}{1}$ **[Option : (A)]**

13. The volume (in cm^3) of the greatest sphere that can be cut off from a cylindrical log of wood of base radius 1 cm and height 5 cm is

(A) $\frac{4}{3}\pi$ (B) $\frac{10}{3}\pi$ (C) 5π (D) $\frac{20}{3}\pi$

Solution

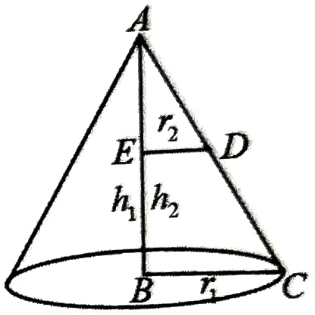
Here, maximum radius of sphere is $1\,cm$.

Volume of the greatest sphere $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi$ **[Option : (A)]**

14. The height and radius of the cone of which the frustum is a part are h_1 units and r_1 units respectively. Height of the frustum is h_2 units and radius of the smaller base is r_2 units. If $h_2 : h_1 = 1:2$ then $r_2 : r_1$ is

(A) 1:3 (B) 1:2 (C) 2:1 (D) 3:1

Solution



Given : $\frac{h_2}{h_1} = \frac{1}{2} \Rightarrow h_2 = k,\ h_1 = 2k$.

From the figure, $AE = h_1 - h_2$.

$\Delta AED \sim \Delta ABC \Rightarrow \frac{AE}{AB} = \frac{ED}{BC}$

$\Rightarrow \frac{h_1 - h_2}{h_1} = \frac{r_2}{r_1} \Rightarrow \frac{2k-k}{2k} = \frac{r_2}{r_1} \Rightarrow \frac{r_2}{r_1} = \frac{1}{2}$ **[Option : (B)]**

15. The ratio of the volumes of a cylinder, a cone and a sphere, if each has the same diameter and same height is

(A) 1:2:3 (B) 2:1:3 (C) 1:3:2 (D) 3:1:2

Solution

Given : $h = 2r$.

Required ratio is

$(\text{Volume of cylinder}) : (\text{Volume of cone}) : (\text{Volume of sphere})$

$\left[\pi r^2 (2r)\right] : \left[\frac{1}{3}\pi r^2 (2r)\right] : \left[\frac{4}{3}\pi r^3\right]$

$2 : \frac{2}{3} : \frac{4}{3}$

i.e., 3:1:2 **[Option : (D)]**

OBJECTIVE TYPE QUESTIONS

EXERCISE 8.5

1. Which of the following is not a measure of dispersion?

(A) Range (B) Standard deviation (C) Arithmetic mean (D) Variance

Solution **[Option:(C)]**

2. The range of the data 8, 8, 8, 8, 8, ... 8 is

(A) 0 (B) 1 (C) 8 (D) 3

Solution

Range $= L - S = 8 - 8 = 0$ **[Option:(A)]**

3. The sum of all deviations of the data from its mean is

(A) Always positive (B) always negative (C) zero (D) non-zero integer

Solution

The sum of all deviations of the data from its mean $\sum(x-\bar{x})=0$ **[Option:(C)]**

4. The mean of 100 observations is 40 and their standard deviation is 3. The sum of squares of all observations is

(A) 40000 (B) 160900 (C) 160000 (D) 30000

Solution

Given : $n=100,\ \bar{x}=40,\ \sigma=3$.

$$\sigma^2=\frac{\sum x^2}{n}-\left(\frac{\sum x}{n}\right)^2 \Rightarrow 9=\frac{\sum x^2}{100}-(40)^2$$

$$\therefore \sum x^2=(1600+9)100=160900$$ **[Option:(B)]**

5. Variance of first 20 natural numbers is

(A) 32.25 (B) 44.25 (C) 33.25 (D) 30

Solution

Variance $\sigma^2=\frac{n^2-1}{12}=\frac{20^2-1}{12}=\frac{399}{12}=33.25$ **[Option:(A)]**

6. The standard deviation of a data is 3. If each value is multiplied by 5 then the new variance is

(A) 3 (B) 15 (C) 5 (D) 225

Solution

Given : $\sigma=3$. The new S.D $=3\times5=15$.

$\therefore$ New Variance $=(15)^2=225$ **[Option:(D)]**

7. If the standard deviation of x, y, z is p then the standard deviation of $3x+5, 3y+5, 3z+5$ is

(A) $3p+5$ (B) $3p$ (C) $p+5$ (D) $9p+15$

Solution

The standard deviation of x, y, z is p.

The $S.D$ of $3x, 3y, 3z$ is $3p$.

The $S.D$ of $3x+5, 3y+5, 3z+5$ is also $3p$.

[Option:(B)]

8. If the mean and coefficient of variation of a data are 4 and 87.5% then the standard deviation is

(A) 3.5 (B) 3 (C) 4.5 (D) 2.5

Solution

Given : $\bar{x}=4,\ C.V=87.5$

$$\Rightarrow \frac{\sigma}{4}\times 100=87.5 \Rightarrow \sigma=\frac{87.5\times 4}{100}=3.5$$

[Option:(A)]

9. Which of the following is incorrect?

(A) $P(A)>1$ (B) $0\le P(A)\le 1$ (C) $P(\phi)=0$ (D) $P(A)+P(\bar{A})=1$

[Option:(A)]

Solution

10. The probability to select a red marble at random from a jar containing p red, q blue and r green marbles is

(A) $\frac{q}{p+q+r}$ (B) $\frac{p}{p+q+r}$ (C) $\frac{p+q}{p+q+r}$ (D) $\frac{p+r}{p+q+r}$

Solution

$$P(\text{a red marble})=\frac{n(\text{red marbles})}{\text{Total number of marbles}}=\frac{p}{p+q+r}$$

[Option:(B)]

11. A page is selected at random from a book. The probability that the digit at units place of the page number chosen is less than 7 is

(A) $\frac{3}{10}$ (B) $\frac{7}{10}$ (C) $\frac{3}{9}$ (D) $\frac{7}{9}$

Solution

Let $S=\{0,1,2,3,\cdots 9\}$; . $n(S)=10$

Let A be the event that the digit at units place of the page number chosen is less than 7.

$\therefore A=\{0,1,2,3,4,5,6\};\ n(A)=7$

Now, $P(A)=\dfrac{n(A)}{n(S)}=\dfrac{7}{10}$ **[Option:(B)]**

12. The probability of getting a job for a person is $\frac{x}{3}$. If the probability of not getting the job is $\frac{2}{3}$ then the value of x is

(A) 2 (B) 1 (C) 3 (D) 1.5

Solution

Given : $P(\text{getting a job})=P(A)=\dfrac{x}{3}$ and $P(\bar{A})=\dfrac{2}{3}$

We know that $P(A)+P(\bar{A})=1 \Rightarrow P(A)=1-P(\bar{A})$

$\Rightarrow \dfrac{x}{3}=1-\dfrac{2}{3}=\dfrac{1}{3}$ or $x=1$ **[Option:(B)]**

13. In a lucky draw contest, 135 tickets of the lucky draw were sold. If the probability of winning is $\frac{1}{9}$, then the number of tickets to be bought, is

(A) 5 (B) 10 (C) 15 (D) 20

Solution

Let A be the event of buying a ticket. Let $n(A)=x$

Given : $n(S)=135,\ P(A)=\dfrac{1}{9} \Rightarrow \dfrac{x}{135}=\dfrac{1}{9}$ or $x=\dfrac{135}{9}=15$ **[Option:(C)]**

14. If a letter is chosen at random from the English alphabets $\{a,b,....,z\}$, then the probability that the letter chosen precedes x

(A) $\frac{12}{13}$ (B) $\frac{1}{13}$ (C) $\frac{23}{26}$ (D) $\frac{3}{26}$

Solution

Let $S=\{a,b,c,\cdots,z\}.\ n(S)=26$

Let A be the event that a letter chosen precedes x. Then $A=\{a,b,c,\cdots w\}$, $n(A)=23$

$\therefore P(A)=\dfrac{n(A)}{n(S)}=\dfrac{23}{26}$ **[Option:(C)]**

15. A purse contains 10 notes of ₹ 2000, 15 notes of ₹ 500, and 25 notes of ₹ 200. One note is drawn at random. The probability that the note is either a ₹500 or ₹ 200 is

(A) $\frac{1}{5}$ (B) $\frac{3}{10}$ (C) $\frac{2}{3}$ (D) $\frac{4}{5}$

Solution

Let A, B, C denote the number of ₹ 2000, ₹ 500, ₹ 200 notes respectively.

$\Rightarrow n(A)=10,\ n(B)=15,\ n(C)=25$ and $n(S)=50$

$$P(\text{either 500 or 200}) = P(B \cup C) = P(B)+P(C)$$

$$= \frac{n(B)}{n(S)}+\frac{n(C)}{n(S)} = \frac{15}{50}+\frac{25}{50} = \frac{40}{50} = \frac{4}{5}$$

[Option:(D)]